

Day 12 – Linux Networking

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1. What is a network interface?

A network interface is the connection point through which a Linux system communicates with a network. It can be a physical device such as Ethernet or Wi-Fi, or a virtual interface.

2. What is the loopback interface?

The loopback interface is a virtual network interface used by the computer to communicate with itself. In Linux it is usually called lo and commonly uses the address 127.0.0.1 for IPv4.

3. What does ip addr display?

The ip addr command displays network interfaces along with their IP addresses, subnet information, interface state, and other address-related details.

4. What is the purpose of ip link?

ip link is used to view and manage network interfaces at the link layer. It can show interface status and can also be used to bring an interface up or down.

5. What does hostname -I display?

hostname -I displays the IP addresses assigned to the current host. It is useful for quickly finding the system's local IP address.

6. What does ip route show?

ip route show displays the system's routing table. It shows how packets are routed to different networks and which gateway/interface is used.

7. What is a default route?

A default route is the route used when no more specific route exists for a destination. It normally sends Internet-bound traffic to the default gateway.

8. What does ping test?

ping tests basic network reachability by sending ICMP Echo Request packets and checking whether Echo Replies are received. It can also show response time and packet loss.

9. Why can a failed ping be misleading?

A failed ping does not always mean the host or network is down. ICMP may be blocked by a firewall, the destination may ignore ping, or there may be a routing or filtering issue.

10. What is DNS?

DNS (Domain Name System) translates human-readable domain names such as google.com into IP addresses that computers can use for network communication.

11. What does getent hosts do?

getent hosts queries the system's configured name-service sources to resolve a hostname into an IP address. It is useful for checking whether hostname resolution is working.

12. What information does ip neigh display?

ip neigh displays the neighbor/ARP cache, showing information such as nearby IP addresses, their link-layer (MAC) addresses, the interface used, and the neighbor state.

13. What is a socket?

A socket is an endpoint used by a program for network communication. It is commonly associated with an IP address, transport protocol, and port number.

14. What does ss -tuln display?

ss -tuln displays listening TCP and UDP sockets without resolving service names or hostnames. It is useful for seeing which local ports are listening.

15. What is the difference between TCP and UDP conceptually?

TCP is connection-oriented and provides reliable, ordered delivery with mechanisms such as acknowledgements and retransmission. UDP is connectionless and has lower overhead but does not guarantee delivery or ordering.

16. What is curl used for?

curl is a command-line tool for transferring data using protocols such as HTTP and HTTPS. It is commonly used to test web services, APIs, URLs, and HTTP responses.

17. What does curl -I request?

curl -I requests only the HTTP response headers (a HEAD request, when supported) instead of downloading the complete response body. It can be used to inspect status codes and headers.

18. What is traceroute used for?

traceroute is used to identify the network hops between a source system and a destination. It helps troubleshoot routing paths, delays, and possible points of failure.

19. Why is ss commonly preferred over netstat on modern Linux?

ss is commonly preferred because it is faster, provides detailed socket information, and is maintained as part of the modern Linux networking toolset. netstat is considered a legacy utility on many Linux systems.

20. Why are Linux networking commands useful in cybersecurity?

They help security professionals understand interfaces, IP addresses, routes, DNS, neighbors, listening services, connections, and network paths. This

supports defensive tasks such as troubleshooting, system monitoring, incident investigation, and security assessment of authorized systems.

Troubleshooting Question

Problem: The Linux system can ping 8.8.8.8, but ping google.com fails because the hostname cannot be resolved.

Answer: This is most likely a DNS/name-resolution problem. Internet connectivity is working at the IP level because 8.8.8.8 is reachable, but the system is unable to translate google.com into an IP address. A suitable first command from today's lab is **getent hosts google.com**, which checks hostname resolution using the system's configured name-service mechanisms.

Practical Evidence – Commands to Capture

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ip addr
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ip route
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ping -c 4 127.0.0.1
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```
getent hosts google.com
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```
ss -tuln
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```
curl -I https://example.com
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